Bob Ma

Senior Machine Learning Engineer · ML Pipelines & MLOps · Data Platforms · Production ML

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WORK RIGHTS NZ Permanent Resident AVAILABLE Immediately

§ 01 SUMMARY

Auckland-based machine learning engineer with a PhD from AUT (machine learning for privacy preservation), an MSc in computer software and theory, 6+ years shipping production ML and 18+ years in software and data engineering. Specialises in the *data pipelines and automation* behind production ML: ETL on AWS (SageMaker, Lambda, Glue, S3) and Snowflake, Spark with Airflow, automated training, evaluation and deployment, and CI/CD (GitHub Actions, Concourse) that ships models behind automated validation, with MLflow tracking and Azure ML endpoints. At Resideo, built VLM-based business assessment for small and medium-sized business (SMB) customers, from front-of-store data through to POS, payment and alerting systems. Delivered across smart-home (Resideo / Honeywell Home), transport (EROAD), retail (Woolworths NZ, StayinFront) and construction (BuildingEstimates); three patents; reviewer for AAAI, ACM TOMM and IEEE journals.

§ 02 CORE SKILLS / STRENGTHS

ML PIPELINES & MLOPS
ETL feeding continuous training, evaluation and deployment; GitHub Actions and Concourse CI/CD; MLflow tracking; Azure ML endpoints; automated validation as a release gate.

DATA PLATFORMS
AWS SageMaker, Lambda, Glue and S3; Spark with Airflow; Snowflake and SQL; vector databases; SAP / ERP integration.

PRODUCTION ENGINEERING & DEVOPS
Python and C++ services (Flask, REST); Docker and Kubernetes; unit, stress and CI/CD testing; Git; Agile / Scrum.

SMB BUSINESS FLOWS
Front-of-store data (shopfront photos, shelf and price-tag images, camera video, POS and inventory) connected to POS, payment, alerting, ticketing and apps; company and client ERP integration.

VLMS, LLMS & AGENTS
VLM business assessments (merchant verification, operating status and risk, shelf compliance, security events) configured per customer; multimodal LLMs via REST APIs; Claude Agent SDK for code review and data processing.

FORECASTING & APPLIED ML
Demand, price and sales forecasting; behaviour modelling (LSTM, k-means); compression, quantisation and distillation.

§ 03 WORK EXPERIENCE

Senior Machine Learning Engineer

Mar 2024 – Oct 2026

Resideo Inc. (Honeywell Home) · NYSE:REZI

- *Production ML: model pipeline, VLM solutions for SMB customers and system integration*
- Designed and optimised multimodal LLM inference (LLaMA3, GPT, CLIP) and integrated model outputs with backend decision systems through REST APIs and edge–cloud pipelines on Azure.
- Owned the end-to-end model pipeline: PyTorch training and optimisation, ONNX export and conversion, experiment and model tracking in MLflow, Docker images released through GitHub Actions, and serving on Azure ML endpoints backed by a vector database.
- Applied the Claude Agent SDK to automated code review in the CI pipeline and to parts of the user-flow data processing, including building the vector-database indexes.

- Led low-level optimisation of transformer-backbone inference on edge devices: quantisation-aware deployment, mixed-precision execution, memory-access optimisation and hardware-aligned scheduling, within strict latency, memory and power budgets.
- Optimised critical kernels and adapted model operators for target chips (Ambarella CV25 / CV72 class) and their embedded inference runtimes.
- For small and medium-sized business (SMB) customers, built front-of-store information pipelines combining shopfront and signage photos, shelf and price-tag images, in-store camera video and store operating data (POS, inventory, pricing).
- Applied vision–language models (VLMs) to business assessments: merchant verification and qualification, operating-status and risk assessment, shelf-display compliance, and security-event analysis with alert summaries.
- Connected assessment results to POS and payment systems, alerting, ticketing and mobile apps; per-customer rules, categories and VLM prompt templates met each SMB customer's custom requirements.
- Built validation and automated CI/CD testing (unit and stress) for production reliability.

Stack: Python · C++ · PyTorch · ONNX · VLMs · Multimodal LLMs · Claude Agent SDK · CLIP · REST APIs · **ML DevOps:** GitHub Actions · Docker · MLflow · Azure ML endpoints · Vector database

Senior Machine Learning Engineer

Mar 2022 — Mar 2024

ERoad Ltd · ASX:ERD

- *ML data platform, ETL pipelines and CI/CD for fleet-scale models*
- Developed and maintained production ETL pipelines on SageMaker, Lambda, Glue, S3, Snowflake and Docker, feeding continuous model training, evaluation and deployment.
- Built Spark processing with Airflow orchestration for fleet itinerary management.
- Used Concourse CI/CD to automate infrastructure provisioning and model deployment, keeping environments reproducible across development, testing and production.
- Processed SMB fleet customers' driving data together with NZTA vehicle-inspection data for vehicle-status and driving-behaviour models.

Stack: Python · C++ · PyTorch · AWS (SageMaker, Lambda, Glue, S3) · PySpark · Airflow · Snowflake · Concourse · Docker · CI/CD

Machine Learning Engineer

Jun 2021 — Feb 2022

StayinFront Ltd

- Processed front-of-store data for retail and SMB clients (shelf and price-tag images, inventory, pricing, shelf life) for sales forecasting, and integrated the results with StayinFront's own ERP and with clients' ERP systems.

Machine Learning Engineer (Contract)

Sep 2020 — Apr 2021

BuildingEstimates.com

- Designed the framework architecture; Python/Flask services on GCP with Kubernetes processing images from Cloud Storage.

Data Engineer

Sep 2017 — Sep 2019

Woolworths NZ · Woolworths Group ASX:WOW

- Designed and developed data pipelines that deliver accurate data to downstream applications.
- Built RESTful API integrations with third-party ERP platforms (SAP portal) for data-warehouse, logistics and payment workflows.

Full Stack Developer · New Image International — Jun 2014 — May 2017

E-commerce and ERP platforms: order processing, customer management, payment-gateway integration.

Software Developer · Huawei Symantec Technology — Dec 2010 — Aug 2013

Led frontend and backend teams; multithreaded Linux C/C++; time-series anomaly detection in spectrum data (Kalman filter, HMM).

Software Developer · Topsec Network Security (002212.SZ) — Oct 2007 — Mar 2010

Linux intrusion detection with kernel-level packet analysis; DDoS detection.

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